Hypergraph Matching Network for Semisupervised Few-Shot Scene Classification of Remote Sensing Images

Jie Geng, *Member, IEEE*, Ran Chen, Bohan Xue, and Wen Jiang

***Abstract*— Semisupervised few-shot learning aims to alleviate the issue of insufficient labeled data with additional unlabeled samples. As for remote sensing images, complex contextual information leads to pseudolabeling with low confidence, which weakens the effect of semisupervised few-shot classification. To solve these issues, a hypergraph matching network is proposed for the semisupervised few-shot scene classification of remote sensing images. Specifically, a hypergraph propagation module is designed to construct a hypergraph network, which can take advantage of adjacent samples with similar semantics and improve the representation ability of class prototypes. Then, a cross-layer prototype matching module is proposed to dynamically match features of different scales and angles, which aims to predict pseudolabels with high confidence. Experimental results on three public remote sensing datasets demonstrate that the proposed method can make effective utilization of additional unlabeled samples to enhance the classification performance of few-shot learning.**

***Index Terms*— Few-shot learning, remote sensing image, scene classification, semisupervised learning.**

I. Introduction

REMOTE sensing is one of the most important technologies to understand the Earth. Scene classification of remote sensing images is essential for analyzing geographical features in the target region. Advances in imaging technology have resulted in higher resolution remote sensing images, while the introduction of computer vision algorithms has allowed an efficient and accurate understanding of high-resolution data, such as scene classification [1], semantic segmentation [2], and target detection [3]. Compared with traditional machine learning approaches, the deep learning model has a stronger recognition ability and is also more suitable for remote sensing classification tasks that excavate complex contextual semantic relations.

Remote sensing image scene classification based on deep neural networks relies on a great number of labeled data. When the training samples are insufficient, the deep model will be overfitting, which affects the final remote sensing scene classification results [4]. Due to the high cost of labeling remote sensing samples and the critical lack of domain-specific remote sensing data, it is of great significance to explore scene classification techniques under the condition of fewer labeled samples. Few-shot learning aims to train the model in order to quickly learn novel tasks through a small number of labeled samples, thus reducing the requirement of numerous samples for deep learning models [5].

The current methods for few-shot learning are mainly data augmentation-based methods, metric learning-based methods, and metalearning-based methods [6]. Data augmentation-based methods solve the issue of insufficient samples by expanding the training data with feature augmentation or generating additional annotations with semisupervised learning. Metric learning-based methods focus on optimizing the metric space of classification so that the same category samples are close to each other and samples of different categories are far away from each other. Metalearning-based models utilize the training set with fewer labels to construct multiple few-shot subtasks, which aims to learn different types of metaknowledge.

However, current methods for few-shot scene classification do not specifically consider the characteristics of remote sensing images, which makes it difficult to capture complex contextual relationships in remote sensing images. Compared with natural color images, remote sensing images generally perform more complex semantics, which leads to unclear category boundaries in the feature space for traditional few-shot scene classification methods. Moreover, the scales and angles of different remote sensing images may be different greatly, and current methods do not have the ability to dynamically match features of different scales and angles.

To address the above issues, a novel hypergraph matching network with semisupervised learning is proposed for the few-shot scene classification of remote sensing images. On the one hand, a novel hypergraph network is proposed to utilize multiple nodes for feature representation, which can represent multiple semantics of a scene in the remote sensing image. It is different from the traditional graph network since it just uses a single node to represent features. On the other hand, a cross-layer prototype matching module is proposed, which leverages iterative clustering to select the most cohesive clusters, where each sample shares similar attributes in scale and angle. In summary, the major contributions of this article are summarized as follows.

1) A hypergraph propagation module is proposed to learn more complex representations, which is able to realize

Received 23 September 2024; revised 20 January 2025; accepted 31 March 2025. Date of publication 7 April 2025; date of current version 24 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271396. *(Corresponding author: Jie Geng.)*

The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: gengjie@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3558639

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

cross-class information interaction without class labels. It takes advantage of adjacent samples with similar semantics and improves the representation ability of class prototypes in the few-shot learning case.

2) A cross-layer prototype matching module is proposed to obtain the prototype set, which enables it to dynamically match features of different scales and angles. It can effectively predict pseudolabels with high confidence, improving the performance of few-shot learning.
3) Experiments on three public remote sensing data demonstrate that our proposed model can make effective utilization of additional unlabeled samples, which improves the classification accuracy of few-shot scene classification.

The remainder of this article is organized as follows. In Section II, related work is presented. In Section III, the proposed method is introduced detailedly. In Section IV, experimental results are shown to demonstrate the effectiveness of our method. Finally, conclusions of this work are summed in Section V.

II. Related Work

A. Scene Classification of Remote Sensing Images

Remote sensing image scene classification analyzes the semantic information of the images to assign the categories. As a basic processing of remote sensing interpretation, scene classification with high efficiency is of great significance for the downstream tasks. According to different feature learning methods, the existing methods can be divided into manual feature extraction-based methods and deep feature extraction-based methods. Manual feature extraction-based methods usually design traditional feature descriptors to extract scene features from high-resolution remote sensing data, which rely on manual expertise and are often accompanied by cumbersome parameter tuning. Thus, manual feature extraction-based methods perform poor generalization ability and robustness [7], [8].

The deep neural network provides a novel approach for remote sensing image interpretation. Penatti et al. [9] introduced convolutional neural networks to the task of remote sensing image classification, which proved that convolutional networks can obtain more accurate classification than low-level descriptor-based methods. Lin et al. [10] developed generative adversarial networks for the scene classification of remote sensing images, which significantly improved the classification precision. Tang et al. [11] proposed an efficient multiscale transformer and cross-level attention model, which is able to mine the underlying contextual information of the images and aggregate the correlation of features between different layers.

As data-driven training approaches, deep neural networks are able to automatically learn rich semantic information from remote sensing images. However, the deep model for scene classification of remote sensing data still faces many problems. On the one hand, a scene region of the remote sensing image is generally composed of multiple targets, whose semantics and scales are different. In the task of scene classification of remote sensing images, relationships between these targets need to be fully considered and interpreted. On the other hand, labeled samples of remote sensing scenes for specific tasks are limited since the cost of sample annotation is high. Deep learning models rely on a great number of labeled data. If the labeled samples are insufficient, the deep model is prone to overfitting, which weakens the final classification effect [12].

B. Few-Shot Image Classification

Few-shot learning aims to learn novel category information from just limited samples [13]. Current few-shot image classification approaches can be mainly classified into three modes, which are data augmentation-based methods, metric learning-based methods, and metalearning-based methods.

Data augmentation-based approaches prefer to improve the generalization of the classification model by augmenting the dataset with fewer labeled samples. Luo et al. [14] use random patches rather than complete images as input units, highlighting foreground semantics and suppressing background semantics in images. Kwitt et al. [15] used fine-grained annotations to estimate feature trajectories for transient attributes and then transformed features of the labeled samples to generate new samples. Zhang et al. [16] segmented the image into foreground and background regions before deep network training and then fused foreground and background features of different images to synthesize pseudoimages, which can expand the training set. Bär et al. [17] proposed a frozen feature augmentation method for few-shot learning, which is proved to improve few-shot performance on several large pretraining datasets.

Metric learning-based methods address the scarcity of labeled data by designing some kind of metrics to calculate sample distance for classification. Kang et al. [18] added a self-attention module and a cross-attention module into the few-shot learning framework, which can convert the basic feature mapping into an autocorrelation tensor and learn the coattention between two image representations. Zhang et al. [19] developed the earth mover's distance (EMD) to compute the similarity between the query set and the support set, which is proved to effectively measure the sample distances, while FRN [20] and DeepEMDv2 [21] are improvements on that work. Chen et al. [22] introduced the semantic information of samples to learn the prior knowledge of each category, thus adjusting the distributions of categories in the feature spaces. Zhang et al. [23] developed a semantic alignment network to obtain class prototypes from semantics and visual features, which can enhance the discrimination of features for few-shot learning. Qin et al. [24] proposed a prototype rectification network to address the domain shifts problem in few-shot hyperspectral image classification, which achieves superior domain alignment. Zheng et al. [25] combined local features with global contextual relationships for few-shot SAR target classification, which is proved to yield effective prototypes and obtain comparable few-shot classification results. In addition, Wang et al. [26] introduced a transformer into few-shot learning in order to achieve superb performance.

Metalearning-based models adopt the training set to construct multiple few-shot subtasks, which can be applied to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

obtain different types of metaknowledge. This metaknowledge includes initial weights of deep network and hyperparameter of the learning rate [27]. Li et al. [28] proposed a meta-SGD model to automatically update the learning rate and the direction of the optimizer. Since metalearning methods are trained on the instance space, they are easily affected by external factors. To address this issue, Zhou et al. [29] developed a deep metalearning model, which can transform the learning goal from concrete examples to abstract concepts. Chen et al. [30] proposed a meta-baseline deep network, which is pretrained with source domain dataset and then trained with a metalearning strategy. Baik et al. [31] designed the metalearning method to learn adaptively generated hyperparameters from metatasks, rather than just learning the initial weights of the network. Ye et al. [32] utilized the teacher–student networks in metalearning to supervise classifiers learned from fewer samples.

All the above few-shot learning networks are able to alleviate the overfitting issue with limited samples. For remote sensing interpretation tasks, there may be numerous unlabeled samples. It is of great significance to assign high-confidence pseudolabels to these unlabeled data and extend the training dataset to enhance the performance of few-shot classification.

C. Semisupervised Few-Shot Image Classification

Semisupervised image classification aims to utilize a great number of samples without labels to extend the training set, thus enhancing the results of classification. With the further development of semisupervised learning, more researchers utilize unlabeled data to alleviate the issue of data deficiency in few-shot learning tasks. Current semisupervised few-shot classification methods can be classified into pseudolabeling-based methods and label propagation-based methods.

In semisupervised image classification, false predictions can weaken the effectiveness of pseudolabels. Pseudolabeling-based methods focus on optimizing or updating the predicted pseudolabels. Huang et al. [33] proposed a pseudoloss confidence metric method, which can map pseudolosses of different few-shot tasks to a uniform metric space, thus learning the prior loss distribution and estimating the confidence of pseudolabels effectively. Wei et al. [34] proposed a novel method for assigning negative labels to support set samples, which simply and effectively improves classification accuracy. Wang et al. [35] proposed an instance credibility inference network focusing on the issue of pseudolabeled noises, which can measure the confidence of pseudolabeled instances based on the sparsity of the parameters. Afterward, the most trustworthy pseudolabeled samples are selected to retrain the linear classifier, and unlabeled samples are iteratively selected for extending the training set. However, pseudolabels with low quality are difficult to avoid, and thus, noisy information cannot be reduced in some cases.

As for the label propagation-based approach, it aims to propagate the label information from labeled nodes to unlabeled nodes by constructing edge connection relationships. Rodríguez et al. [36] proposed an unsupervised embedding propagation network to address the issue of distribution bias between the training set and the unlabeled set, which can build a relational graph by calculating the similarity between features. Liu et al. [37] proposed a transductive propagation network, which is able to propagate the label information by constructing a graph module. However, the above methods only consider the links between different samples in a single mode, ignoring the associations between different layer features. Therefore, in our proposed method, the correlation information between different layer features is learned through a multimode graph propagation network, so as to obtain the pseudolabels with higher confidence.

III. Methodology

A. Overview

As for scene classification of remote sensing data, it is difficult to acquire a great number of annotation data. In order to solve the issues of insufficient labeled samples and complex contextual semantics of remote sensing images, a hypergraph matching network is proposed for semisupervised few-shot scene classification, which includes a hypergraph propagation module and a cross-layer prototype matching module. The hypergraph propagation module utilizes a multilayer self-attention mechanism to extract features and constructs a hypergraph in order to generate feature propagation results. Afterward, the support set is expanded by unlabeled samples through a cross-layer prototype matching module. The proposed framework can be seen in Fig. 1.

B. Task Definition

Few-shot classification aims to classify samples with quite limited labeled data. It includes several different task modes that report as follows. Both inductive few-shot learning and semisupervised few-shot learning models are trained on the train set D_{train} and tested on the test set D_{test} with limited samples. As for few-shot learning scenarios, the classes of D_{train} and the classes of D_{test} should not be overlapped, which means $C_{\text{train}} \cap C_{\text{test}} = \emptyset$. For both D_{train} and D_{test}, the few-shot learning processes are organized by episode as metalearning. The whole dataset is divided into many episodes, and each episode represents a learning subtask. As for inductive few-shot learning with N-way K-shot, the dataset in an episode consists of support set $\{x^s_{n,k}, y^s_{n,k} | n = 1, \ldots, N; k = 1, \ldots, K\}$ and a query set $\{x^q_{n,m} | n = 1, \ldots, N; m = 1, \ldots, M\}$, where $x^s_{n,k}$ denotes the kth sample from the nth class and $y^s_{n,k}$ stands for the corresponding label. The inductive learning model f_θ utilizes fewer labeled samples on the support set to adapt unseen classes and then classify the samples on the query set. It can be written as

$$y^q_{n,m} \leftarrow f_\theta\left(x^q_{n,m} \middle| \left\{x^s_{n,k}, y^s_{n,k} \middle| n = 1, \ldots, N; k = 1, \ldots, K\right\}\right). \tag{1}$$

Different from inductive learning, semisupervised few-shot learning takes some unlabeled samples to expand the support set. As for semisupervised few-shot learning with N-way K-shot U-unlabel, the unlabeled set $\{x^{un}_{n,u} | n = 1, \ldots, N; u = 1, \ldots, U\}$ and the support set $\{x^s_{n,k}, y^s_{n,k} | n = 1, \ldots, N; k =$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

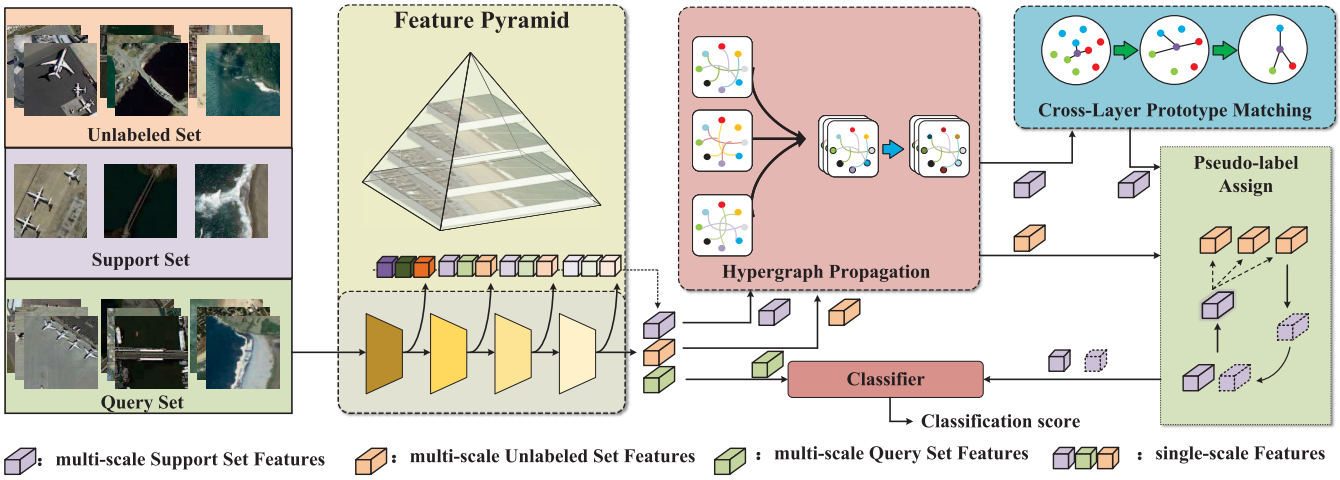


Fig. 1. Proposed framework of the hypergraph matching network. First, the feature pyramid module extracts multiscale features of remote sensing images and obtains the learned feature sets. Then, the support feature sets and unlabeled feature sets are imported into the hypergraph propagation module to optimize the feature representation. After that, the cross-layer prototype matching module is developed to calculate the class prototypes, which are utilized to predict the pseudolabels of the unlabeled data. Finally, the pseudolabeled samples are supplemented into the support set, which is adopted for the classification of query set samples.

$1, \ldots, K\}$ are used to adapt unseen classes and then classify the samples on the query set $\{x^q_{n,m} | n = 1, \ldots, N; m = 1, \ldots, M\}$. It can be defined as

$$y^q_{n,m} \leftarrow f_\theta\left(x^q_{n,m} \middle| \left\{x^s_{n,k}, y^s_{n,k} \middle| n = 1, \ldots, N; k = 1, \ldots, K\right\}; \left\{x^{un}_{n,u} \middle| n = 1, \ldots, N; u = 1, \ldots, U\right\}\right). \quad (2)$$

Normally, the sample number of support set K is much smaller than the number of unlabeled set U in the few-shot scenario. Besides, there is a special semisupervised few-shot learning called transductive few-shot learning, which merges the query set with the unlabeled set. It can be defined as

$$y^q_{n,m} \leftarrow f_\theta\left(x^q_{n,m} \middle| \left\{x^s_{n,k}, y^s_{n,k} \middle| n = 1, \ldots, N; k = 1, \ldots, K\right\}; \left\{x^q_{n,m} \middle| n = 1, \ldots, N; m = 1, \ldots, M\right\}\right). \quad (3)$$

C. Hypergraph Propagation Module

As for scene classification of remote sensing data, feature representation can be regarded as multipatterns. For example, the "airport" scene is generally represented by combining multiple scenes such as "airplanes," "runways," and "buildings." Traditional methods cannot understand the connection between various scenes effectively. To solve the above issues, a hypergraph propagation module is developed to learn more complex representations of hidden relationships between images through multimode feature propagation, where the specific process is depicted in Fig. 2.

1) Pretraining Process: During the pretraining process, the feature pyramid and self-attention mechanism are added into the feature extractor $f(\cdot)$ in order to extract features $F = \{F_h\}_{h=1,\ldots,H}$ from the image X, i.e., $F = f(X)$, where H denotes the number of modalities. To further enhance the discrimination of features, supervised contrastive learning is used, where features of the same category are set as positive sample pairs and others as negative sample pairs. The specific

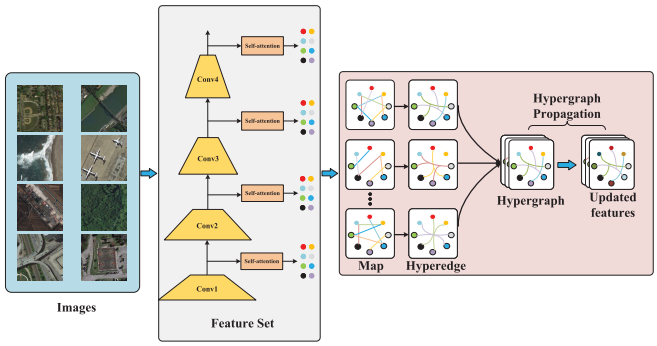


Fig. 2. Structure of the hypergraph propagation module. Feature sets output by different layers correspond to different hyperedges, which are utilized to construct the hypergraph.

formula can be written as

$$\mathcal{L}^h = -\sum_{i=1}^{2n} \frac{1}{2n_{y_j} - 1} \sum_{j=1}^{2n} \mathbb{I}_{[i \neq j]} \cdot \mathbb{I}_{[y_i = y_j]} \times \log \frac{\exp(F_{h,i} \cdot F_{h,j})/\tau}{\sum_{k=1}^{2n} \mathbb{I}_{[i \neq k]} \exp(F_{h,i} \cdot F_{h,k}/\tau)} \quad (4)$$

$$\mathcal{L} = \sum_{h=1}^{H} \mathcal{L}^h \quad (5)$$

where $F_{h,i}$ denotes the hth feature for the anchor point and $\mathcal{L}^h$ denotes the supervised contrastive learning loss for the hth feature. $F_{h,j}$ denotes the positive sample corresponding to $F_{h,i}$; on the contrary, $F_{h,k}$ forms a negative sample pair with $F_{h,i}$. τ stands for the temperature coefficient, and if τ is smaller, the model will focus more on difficult negative samples. By constructing multilayer supervised comparative learning, the model can exploit the similarity of the same category samples within different layers, as well as the differences between different categories, so as to yield more discriminative features.

2) Hypergraph Propagation: Graph-based models can construct similarity matrices through the joint relationship

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

between labeled and unlabeled data, which has achieved remarkable results for semisupervised few-shot classification. However, the traditional method only considers the correlation between single-mode features, which cannot solve the complex semantic features of remote sensing images. Thus, a hypergraph network is developed for semisupervised few-shot tasks, in which the hypergraph structures consider hidden representations of higher level features to encode the correlations, thereby learning more complex relationships of remote sensing images.

In the testing process, for each few-shot task, the trained feature extractor is developed to extract the support set feature $F^s \in R(N \times K, H, d)$, the query set feature $F^Q \in R(N \times Q, H, d)$, and the unlabeled set feature $F^U \in R(N \times U, H, d)$, respectively. The hypergraph is constructed jointly using the support set feature and the unlabeled set feature $F^Z \in R(N \times (K+U), H, d)$, where $Z = S \cup U$ and d denotes the feature dimensions. In order to obtain the hyperedge E_h corresponding to the hth mode feature, the correlation matrix W is calculated using node $V = F_h^z$, where the formula is given as follows:

$$W_{i,j} = \frac{F_{i,h}^Z \cdot \left(F_{j,h}^Z\right)^T}{\left\|F_{i,h}^Z\right\|_2 \left\|\left(F_{j,h}^Z\right)^T\right\|_2} \tag{6}$$

where the cosine similarity between two features is calculated to obtain the correlation matrix and $\|\cdot\|_2$ represents the Euclidean norm. After that, each node is set as the center, and the most similar m nodes are selected to construct the edge $E(|E| = |V| = N \times (K+U))$, thus obtaining the hyperedge $P \in R(|E|, |V|)$, where the formula can be written as follows:

$$P_{i,j} = \begin{cases} 1, & \text{if } F_{i,h}^Z \in \text{neighbor}\left(F_{j,h}^Z\right) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where neighbor$(\cdot)$ denotes the set of the most similar m features to the corresponding feature. Similarly, the corresponding hyperedges are obtained for other layer features. Furthermore, the hyperedge $P^{\text{fuse}} \in R(|V| \times H|E|)$ is obtained by concatenation operation considering the relation between the multilayer features. According to [38], hypergraph G can be constructed by the following formula:

$$G = D_v^{-1/2} P D_e^{-1/2} P^T D_v^{-1/2} \tag{8}$$

where $D_v[i,i] = \sum_j P[i,j]$ denotes the point-degree diagonal matrix and $D_e[j,j] = \sum_i P[i,j]$ stands for the edge-degree diagonal matrix.

In this work, the hypergraph network is introduced for few-shot feature propagation in semisupervised conditions, where feature propagation is performed using hypergraph convolution. Its advantage is that it does not require an additional training process, thus effectively avoiding the overfitting problem caused by too few samples in the novel category. After yielding the hypergraph G, feature propagation is adopted to obtain the hypergraph propagated feature $\widetilde{F^z}$, where the formula is written as

$$\widetilde{F^Z} = (\alpha I + E)^k F^Z \tag{9}$$

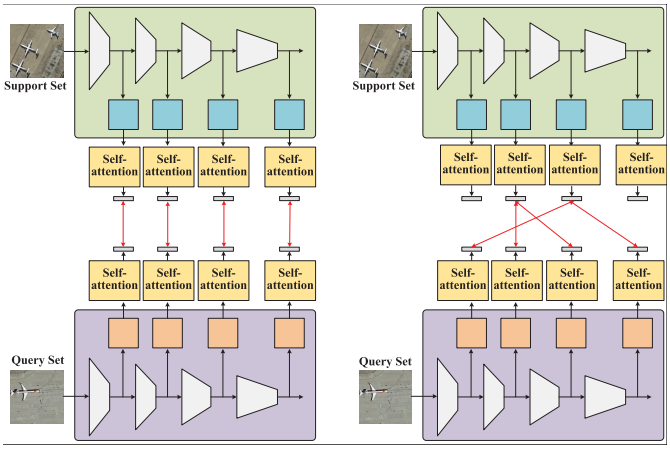


Fig. 3. Comparison of two feature set metrics. (Left) Feature set metric with fixed matching, where each mode feature points to the object in parallel. (Right) Feature set metric with adaptive matching, designed to match the most similar objects for each mode feature.

where I stands for the identity matrix and α is the hyperparameter to control the balance between itself and its neighbors. k denotes the degree of feature propagation. When k is larger, the feature propagation is deeper, and the node will consider more information from the neighboring nodes. Through hypergraph propagation, it learns the hidden associations between features at different scales from the images, which can reduce the feature mismatch problem caused by the complex contextual semantics of remote sensing images.

After obtaining the feature $\widetilde{F^Z}$, the support feature $\widetilde{F^S}$ and the unlabeled sample feature $\widetilde{F^U}$ after feature propagation can be generated. Traditional methods obtain the prototype of labeled samples by taking the mean of features, which is utilized as the classifier weights to obtain pseudolabels. Here, two different similarity measure methods to calculate the distance between feature sets are analyzed, which is shown in Fig. 3. These two similarity metrics can be written as follows:

$$d(F_a, F_b) = \sum_{i=1}^{H} \text{Sim}(F_{a,i}, F_{b,i}) \tag{10}$$

$$d(F_a, F_b) = \sum_{i=1}^{H} \max_{j=1}^{H} \text{Sim}(F_{a,i}, F_{b,j}) \tag{11}$$

where H represents the number of feature modes. Equation (10) represents the sum of cosine similarity corresponding to the layer subfeatures, while (11) represents the sum of maximum cosine similarity across the layer subfeatures. Comparisons of these two similarity metrics are discussed in the experiments, which are reported in the ablation studies with different set metrics. It can be seen that cosine similarity across the layer subfeatures yields superior results, and thus, this metric is adopted for feature measure.

D. Cross-Layer Prototype Matching Module

Due to discrepancies in resolution, shooting angle, and so on between different remote sensing images, it is not an optimal matching method for corresponding layer features in the same scene. Thus, simply calculating the average of corresponding layer features to yield the prototype set is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

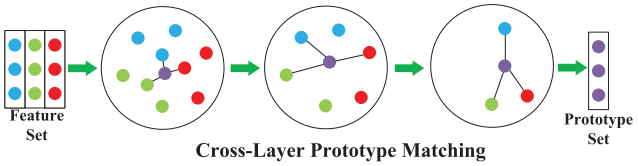


Fig. 4. Structure of the cross-layer prototype matching module. For each iteration, most similar features are searched, and the mean value is calculated as the class prototype for this iteration.

unreasonable. To address this problem, a cross-layer prototype matching module is proposed in our framework, where the specific structure is presented in Fig. 4.

For the feature from each support set of the same class, K most similar features are searched. This module first takes the feature from each support set of the same class in order to obtain the largest similarity sum of K features. After that, these K features are averaged as the feature of the prototype set, which is removed from the support feature set at the same time. This process is iterated until no feature exists in the support feature set, thus obtaining the prototype set $p_n = \{p_{n,h}\}_{h=1,\ldots,H}$, whose formula is written as follows:

$$p_{n,1} = \frac{1}{K}\sum_{k=1}^{K} \widetilde{F}^{S}_{n,k,\sigma(k)} \tag{12}$$

where n stands for the class index and σ stands for the index set of K most similar features, which is yielded as follows:

$$\sigma = \arg\max \sum_{0\le i<j\le K} \mathrm{Sim}\left(z_{i,\sigma(i)}, z_{j,\sigma(j)}\right) \tag{13}$$

where Sim denotes cosine similarity and z denotes the subfeature. After obtaining the prototype set p, for an unlabeled feature set $F_i^U = \{F_{i,h}^U\}_{h=1,\ldots,H}$, the similarity between these two sets is calculated as follows:

$$S_{n,i} = \sum_{h=1}^{H} \max_{m=1}^{H} \mathrm{Sim}\left(F_{i,h}^U, p_{n,m}\right) \tag{14}$$

where $S_{n,i}$ stands for the similarity between the ith unlabeled sample and the nth prototype. On this basis, the pseudolabels $\widetilde{y}^U$ of unlabeled samples can be predicted. Afterward, unlabeled samples and support samples of the same category are merged to expand the labeling information.

IV. Experiments

In this section, the specific division of the datasets and the setting of hyperparameters are reported first. Then, compared results are reported to illustrate the efficiency of our method. Finally, the contributions of different modules and parameters are verified by ablation studies.

A. Dataset

1) NWPU-RESISC45 Dataset: The NWPU-RESISC45 dataset is composed of 31 500 images with the resolution of 256 × 256 pixels [39]. It consists of 45 scene categories and 700 images of each category, including airplane, parking lot, railroad, and so on. These samples are randomly divided into 25 classes as the training set, ten classes as the validation set, and ten classes as the test set. Some sample examples of the NWPU-RESISC45 dataset are shown in Fig. 5.

2) AID Dataset: The AID dataset is composed of 30 categories of scene images [40], in which each category includes about 200 to 400 images, totaling 10 000 images overall. 16 categories are chosen for training, seven categories are chosen for validation, and the remaining seven categories are chosen for testing. Some sample examples of the AID dataset are shown in Fig. 6.

3) UCM Dataset: The UCM dataset consists of 2100 images with 21 categories [41], where ten, five, and six categories are selected as training, validation, and test sets, respectively. Some sample examples of the UCM dataset are shown in Fig. 7.

B. Experimental Setup

1) Network Architectures: In the proposed framework, ResNet-12 is chosen as the backbone network, which includes four residual blocks. Among them, each residual block consists of three 3 × 3 convolutional layers, the BatchNorm layer, and the LeakyReLU layer. After each residual block, a 2 × 2 max-pooling layer is used to reshape the size of features. During pretraining, we pass the last layer of features through a global average pooling layer and employ a logistic regression classifier for classification.

2) Hyperparameters: During the pretraining of our model, SGD is chosen as the optimizer with an initial learning rate of 0.5 and a weight decay coefficient of 0.0001. It contains 200 epochs with a learning rate decaying at the 180th epoch, and the decay multiplier is 0.1. The batch size is set to 128. The temperature coefficient τ in the loss function is set to 10.0. Moreover, hyperparameters of the proposed method m, α, and k are set to 6, 0.9, and 3, respectively.

3) Compared Methods: To prove the effectiveness of the proposed hypergraph matching network, several advanced methods have been applied for comparisons. The compared method includes Meta-baseline [30], DLA-MatchNet [42], SCL-MLNet [43], TSC [44], TPN [37], EPNet [36], ICI [35], PLCM [33], IDLN [45], cluster-FSL [46], and iLPC [47]. Among them, Meta-baseline, DLA-MatchNet, SCL-MLNet, and TSC are inductive few-shot classification methods; IDLN and TPN are transductive few-shot learning methods; and EPNet, ICI, PLCM, cluster-FSL, and iLPC are semisupervised methods. For more comprehensive comparisons, the proposed method is also conducted in a transductive setting. For semisupervised few-shot learning, both 5-way 1-shot 30-unlabel and 5-way 5-shot 50-unlabel settings are applied in the experiments.

4) More Details: All the experiments are conducted on four NVIDIA RTX3090 GPUs with 24 GB, and the experimental results are calculated by averaging the five results.

C. Experimental Results

Table I reports the experimental results on the NWPU-RESISC45 dataset and the AID dataset. It is clear that classification accuracies of semisupervised methods are superior to transductive learning methods and inductive learning methods, which indicates that the overfitting problem can be effectively alleviated by introducing unlabeled samples. Compared with other semisupervised methods, our proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS ON NWPU-RESISC45 AND AID DATASETS UNDER 5-WAY SETTING. THE BEST RESULTS HAVE BEEN HIGHLIGHTED

Setting	Method	Backbone	NWPU-RESISC45		AID	
			1-shot	5-shot	1-shot	5-shot
Inductive	DLA-MatchNet [42]	-	68.80 ± 0.70	81.63 ± 0.46	57.21 ± 0.82	73.45 ± 0.61
	SCL-MLNet [43]	-	62.21 ± 1.12	80.86 ± 0.76	59.46 ± 0.96	76.31 ± 0.68
	TSC [44]	ResNet-12	73.26 ± 0.15	84.62 ± 0.35	68.21 ± 0.83	81.72 ± 0.51
	Meta-baseline [30]	ResNet-12	65.33 ± 0.69	81.71 ± 0.44	63.47 ± 0.90	77.33 ± 0.58
Transductive	Meta-baseline	ResNet-12	69.49 ± 0.88	84.42 ± 0.48	66.74 ± 0.87	81.28 ± 0.54
	TPN [37]	4 CONV	65.46	77.21	66.61	78.66
	EPNet [36]	ResNet-12	69.76 ± 0.93	84.00 ± 0.49	67.09 ± 0.97	79.55 ± 0.57
	IDLN [45]	ResNet-12	75.25 ± 0.75	84.67 ± 0.23	68.65 ± 0.46	80.58 ± 0.27
	HMNet (ours)	ResNet-12	79.91 ± 0.89	88.43 ± 0.43	75.32 ± 0.90	85.16 ± 0.46
Semi-supervised	Meta-baseline	ResNet-12	75.83 ± 0.93	86.33 ± 0.43	71.13 ± 0.96	82.90 ± 0.49
	EPNet	ResNet-12	76.17 ± 0.96	86.58 ± 0.51	71.78 ± 0.95	82.23 ± 0.57
	ICI [35]	ResNet-12	77.93 ± 1.12	83.99 ± 0.70	70.52 ± 1.20	82.00 ± 0.75
	cluster-FSL [46]	ResNet-12	82.67 ± 0.80	87.83 ± 0.44	75.33 ± 0.83	85.82 ± 0.42
	iLPC [47]	ResNet-12	79.22 ± 0.97	85.26 ± 0.52	72.92 ± 1.00	84.50 ± 0.58
	PLCM [33]	ResNet-12	81.93 ± 1.03	87.02 ± 0.64	74.68 ± 1.07	85.17 ± 0.70
	HMNet (ours)	ResNet-12	**84.54 ± 0.88**	**89.80 ± 0.42**	**77.85 ± 0.91**	**87.68 ± 0.41**

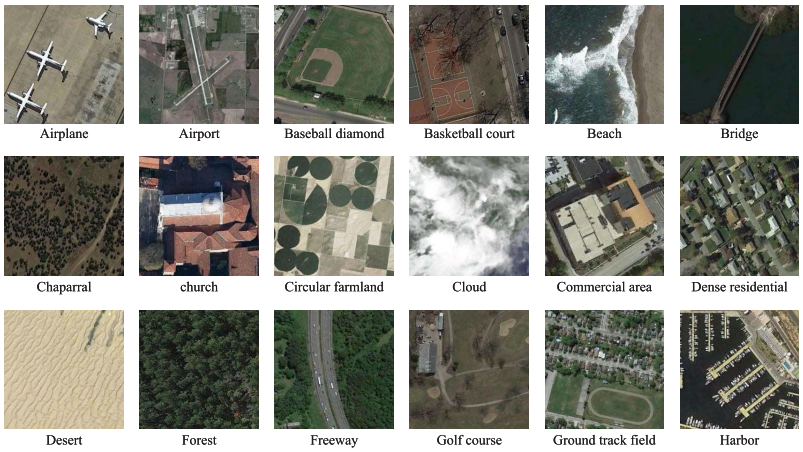


Fig. 5. Sample examples of the NWPU-RESISC45 dataset.



Fig. 6. Sample examples of the AID dataset.

hypergraph matching network achieves great improvements, which yields 8.71% and 8.37% improvements compared to Meta-baseline and EPNet in 1-shot case, respectively. Compared with recent state-of-the-art models, our method still achieves excellent results. Compared with the cluster-FSL model, the hypergraph matching network improves the classification accuracy by 1.87% and 1.97% in the 1-shot case and the 5-shot case, respectively. Among them, both EPNet and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Sample examples of the UCM dataset.

TABLE II
RESULTS OF COMPARED EXPERIMENTS ON THE UCM DATASET UNDER 5-WAY SETTING

Setting	Method	UCM	
		1-shot	5-shot
Inductive	DLA-MatchNet [42]	53.76 ± 0.62	63.01 ± 0.51
	SCL-MLNet [43]	51.37 ± 0.79	68.09 ± 0.92
	TSC [44]	55.11 ± 0.68	69.20 ± 0.64
	Meta-baseline [30]	54.22 ± 0.76	67.27 ± 0.60
Transductive	Meta-baseline	56.46 ± 0.73	70.44 ± 0.61
	TPN [37]	61.77	74.81
	EPNet [36]	54.64 ± 0.78	68.12 ± 0.65
	IDLN [45]	60.09 ± 0.44	73.27 ± 0.25
	HMNet (ours)	63.76 ± 0.86	76.05 ± 0.51
Semi-supervised	Meta-baseline	58.83 ± 0.77	72.13 ± 0.62
	EPNet	57.03 ± 0.72	71.34 ± 0.63
	ICI [35]	56.91 ± 1.20	70.25 ± 0.76
	cluster-FSL [46]	64.07 ± 0.85	75.82 ± 0.60
	iLPC [47]	60.73 ± 0.98	73.79 ± 0.57
	PLCM [33]	62.89 ± 1.10	75.01 ± 0.70
	HMNet (ours)	**66.38 ± 0.85**	**76.69 ± 0.51**

TPN obtain the pseudolabels through information propagation, and the compared results prove that our proposed network is more effective in digging out the associations between features in different layers and, thus, achieves more reliable label information propagation.

From experimental results on the AID dataset, it can be seen that our method achieves optimal classification precision in both transductive and semisupervised settings. Compared to PLCM, our method improves in both 5-way 1-shot 30-unlabel case and 5-way 5-shot 50-unlabel case by 3.17% and 2.51% in semisupervised settings, respectively. It indicates that our proposed approach is conducive to making better utilization of the limited labeling information for classification.

For the UCM dataset, compared results are shown in Table II. In a transductive setting, the hypergraph matching network improves 3.67% and 2.78% in accuracies compared to IDLN under 1-shot 30-unlabel and 5-shot 50-unlabel cases, respectively. Compared with other methods, our proposed model also yields the best classification results, which further testifies to its efficacy and robustness.

In summary, our proposed hypergraph matching network achieves excellent classification results on all three datasets.

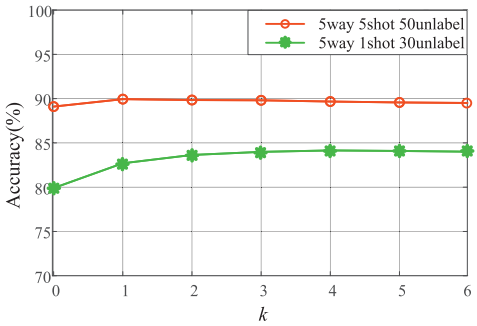


Fig. 8. Classification accuracy with different parameters k's.

It is proved that the proposed model can effectively achieve the information propagation between different layer features, so as to obtain higher quality pseudolabels. Moreover, the cross-layer prototype matching module is able to compute the prototype sets of labeled samples more accurately.

D. Ablation Studies

1) Discussions of Hypergraph Propagation Parameters: In (9), the hyperparameter k represents the degree of feature propagation. A larger k indicates a deeper interaction between different samples, and the node can consider more information from neighboring nodes, which results in feature oversmoothing. On the contrary, too small k cannot learn enough information from neighboring nodes. Therefore, in this section, the accuracy of hypergraph classification is analyzed with different values of k.

In Fig. 8, the red folded line stands for the accuracy in the 5-way 5-shot 50-unlabel condition, and the green folded line stands for the accuracy in the 5-way 1-shot 30-unlabel condition. With the increase in k, classification accuracy performs a trend of rising and then decreasing. As can be seen from the figure, in the 5-way 1-shot 30-unlabel case, the model achieves the optimal results when k is equal to 3. In the 5-way 5-shot 50-unlabel case, the model yields the best accuracy when k is equal to 1. It can be seen that the sample information is more sufficient in the 5-shot condition, resulting in the feature propagation process does not require more information on other samples compared to the 1-shot condition. Considering that classification accuracies of $k = 1$ and $k = 3$ are very close in the 5-shot case, k is set to 3 in all subsequent experiments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDIES OF DIFFERENT MODULES

Backbone	Feature set	Hypergraph	Prototype matching	NWPU-RESISC45	AID	UCM
✓	-	-	-	87.00 ± 0.40	83.43 ± 0.52	73.97 ± 0.51
✓	✓	-	-	88.10 ± 0.41	85.71 ± 0.43	75.65 ± 0.49
✓	-	✓	-	89.55 ± 0.41	86.19 ± 0.44	76.42 ± 0.49
✓	-	✓	✓	88.51 ± 0.39	86.51 ± 0.45	75.92 ± 0.50
✓	✓	✓	✓	89.80 ± 0.42	87.68 ± 0.41	76.73 ± 0.51

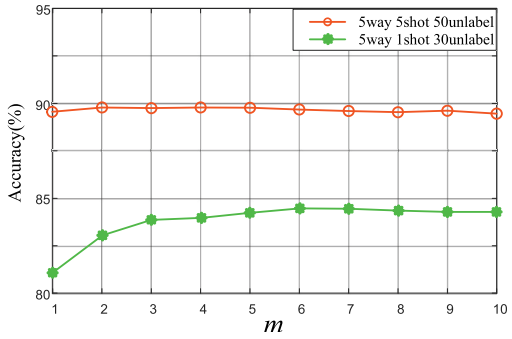


Fig. 9. Classification accuracy with different parameters m's.

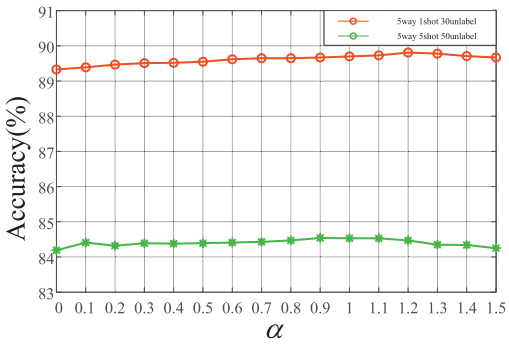


Fig. 10. Classification accuracy with different parameters α's.

Based on (7), for each node, m nodes with the largest expected similarities are selected for constructing hyperedges, which, in turn, constructs the hypergraph network and performs feature propagation. Here, the hyperparameter m is used to control the channel number for information transfer between nodes in the graph network. If m is too large, the graph network is able to combine more nodes for information propagation, but it tends to introduce noise. If m is too small, it is difficult to learn between different nodes through the graph network. Thus, the setting of m is discussed as follows.

As shown in Fig. 9, the parameter m does not have a significant effect on classification accuracy in the 5-way 5-shot 50-unlabel condition. In the 5-way 1-shot 30-unlabel condition, classification accuracy shows a clear tendency to increase and then decrease with the increase in m, which yields the highest accuracy when $m = 6$. It indicates that the similarity relationship can be effectively learned through feature propagation with suitable m. Thus, the hyperparameter m is set to 6 in the subsequent experiments.

Moreover, in our proposed hypergraph matching network, the hyperparameter α is used to control the balance between each node and the corresponding similar nodes. The discussion of α is shown in Fig. 10. As can be seen from the figure, the parameter α has no obvious effect on the classification accuracy. Under the two different conditions, as the parameter α increases, the classification accuracy first increases and then decreases, and reaches the highest when α is set to 0.9 and 1.2, respectively. It indicates that nodes retaining too much of their own information or learning too much of other nodes' information will reduce the final classification accuracy. In our work, α is set to 0.9 for subsequent experiments.

2) Ablation Studies With Different Modules: The overall framework of the hypergraph matching network consists of two modules, including the hypergraph propagation module and the cross-layer prototype matching module. The hypergraph propagation module is developed to extract features and then construct a hypergraph network. The cross-layer prototype matching module is developed for semisupervised learning to obtain pseudolabels from unlabeled samples. In addition, the "Feature set" module stands for the feature set metric with adaptive matching. Ablation experiments are conducted to validate the effectiveness of each module. Experimental results of ablation studies under 5-way 5-shot conditions on three datasets are reported in Table III, respectively.

It is observed that "Backbone + Feature set" achieves higher classification accuracy compared with "Backbone," which indicates that the feature set matching method can alleviate the issue of semantic complexity in remote sensing images. Classification results of "Backbone + Hypergraph" are also higher than that of backbone, whose major reason is that the hypergraph propagation module can mine the hidden representations between different layers of the features. It can be seen that "Backbone + Hypergraph + Prototype matching" does not show significant performance improvement. Our proposed method, which is "Backbone + Feature set + Hypergraph + Prototype matching," yields the highest few-shot classification performance. It is observed that combining "Feature set" and "Prototype matching" has a great contribution to few-shot classification since the feature set matching method can provide multiscale and multiangle features, and the proposed cross-layer prototype matching module is able to dynamically match features of different scales and angles. Therefore, two core modules of the proposed method, the hypergraph network and the cross-layer prototype matching module, are both effective for improving the classification accuracy of few-shot learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDIES WITH DIFFERENT SET METRICS

Metric method	5-way 1-shot	5-way 5-shot
Feature set + Eq. (10)	79.23 ± 0.87	87.66 ± 0.43
Feature set + Eq. (11)	79.90 ± 0.88	88.10 ± 0.42
Ours + Eq. (10)	84.04 ± 0.89	89.16 ± 0.70
Ours + Eq. (11)	84.54 ± 0.88	89.80 ± 0.69

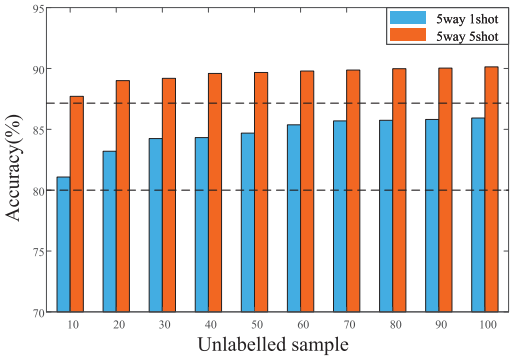


Fig. 11. Classification accuracy with different unlabeled sample numbers.

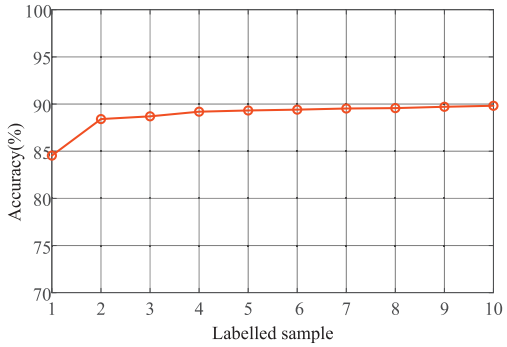


Fig. 12. Classification accuracy with different labeled sample numbers.

3) Ablation Studies With Different Set Metrics: In the hypergraph propagation module, the hypergraph is constructed by feature sets, and information is propagated between labeled and unlabeled samples with two feature set metrics. Classification accuracies with different metrics on the NWPU-RESISC45 dataset are reported in the following table. It can be seen from Table IV that our cross-layer set matching method yields higher classification accuracy than the corresponding layer set matching method in both settings. It indicates that the cross-layer set matching is superior to the corresponding layer set matching for feature measure.

4) Ablation Studies With Different Sample Numbers: Comparisons of classification performance with different numbers of unlabeled samples on the NWPU-RESISC45 data are shown in Fig. 11. The red bars denote the classification results in the 5-way 5-shot condition, and the blue bars denote the classification results in the 5-way 1-shot condition. It is seen from the figure that classification accuracy shows a positive correlation with the number of unlabeled samples, which grows rapidly between ten-unlabel and 50-unlabel samples. It indicates that the proposed model is efficient in learning labeling knowledge from additional unlabeled data.

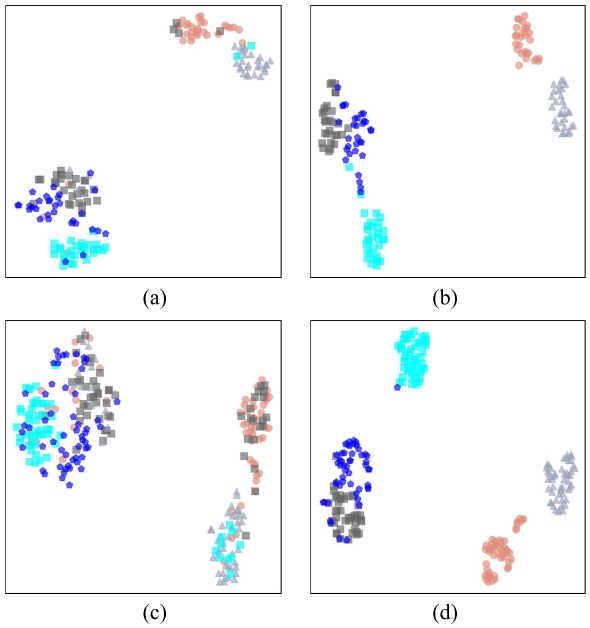


Fig. 13. t-SNE visualization of features on the NWPU-RESISC45 dataset, where features obtained from only the backbone are compared with our proposed method. (a) Feature visualization of backbone in the 5-way 1-shot 30-unlabel case, (b) feature visualization of the proposed method in the 5-way 1-shot 30-unlabel case, (c) feature visualization of backbone in the 5-way 5-shot 50-unlabel case, and (d) feature visualization of the proposed method in the 5-way 5-shot 50-unlabel case.

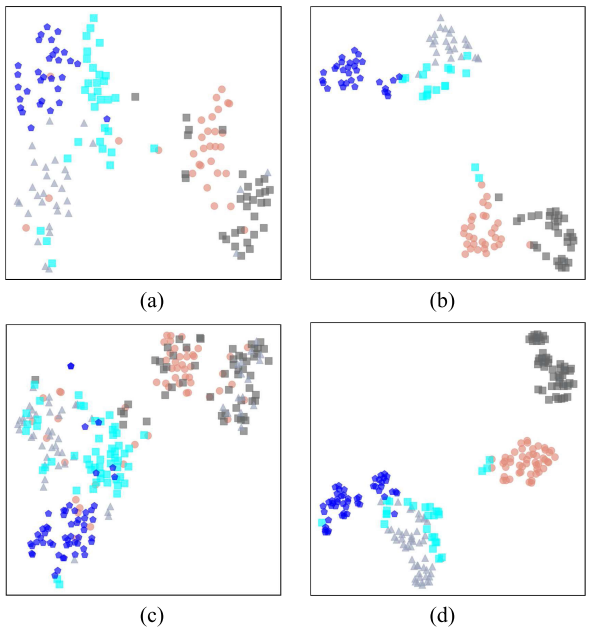


Fig. 14. t-SNE visualization of features on the UCM dataset, where features obtained from only backbone are compared with our proposed method. (a) Feature visualization of backbone in the 5-way 1-shot 30-unlabel case, (b) feature visualization of the proposed method in the 5-way 1-shot 30-unlabel case, (c) feature visualization of backbone in the 5-way 5-shot 50-unlabel case, and (d) feature visualization of the proposed method in the 5-way 5-shot 50-unlabel case.

In addition, classification accuracies with different labeled samples are shown in Fig. 12. For fair comparisons, unlabeled samples are uniformly set to 30. Classification accuracy increases with the increase in labeled samples. It suggests that a smaller number of labeled samples are more likely to cause the model to misclassify.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

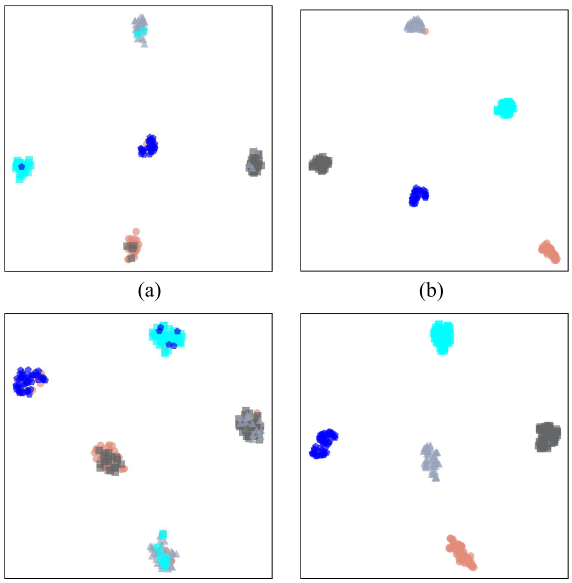


Fig. 15. t-SNE visualization of features on the AID dataset, where features obtained from only backbone are compared with our proposed method. (a) Feature visualization of backbone in the 5-way 1-shot 30-unlabel case, (b) feature visualization of the proposed method in the 5-way 1-shot 30-unlabel case, (c) feature visualization of backbone in the 5-way 5-shot 50-unlabel case, and (d) feature visualization of the proposed method in the 5-way 5-shot 50-unlabel case.

E. Visualization

Visual comparisons of features learned by our proposed method and only backbone on the NWPU-RESISC45 dataset are depicted in Fig. 13. Here, the backbone means no use of the proposed hypergraph algorithm. From the visual comparisons, it can be found that our hypergraph matching network effectively demarcates the boundaries of different clusters and makes features of the same class closer. It proves that the proposed method can enhance feature discrimination.

To further illustrate effectiveness, visual comparisons of the UCM dataset and the AID dataset are shown in Figs. 14 and 15, respectively. It is observed that our hypergraph structure significantly optimizes the feature representation both in 1- and 5-shot cases, where features from our hypergraph matching network show better clustering effects. Especially on the AID dataset, our hypergraph structure can reduce the number of samples located in the wrong clusters, which means our method can reduce the feature confusion between different categories. Therefore, feature visualizations demonstrate that our proposed hypergraph matching network is able to learn more effective features.

V. Conclusion

In this article, a novel hypergraph matching network is proposed for semisupervised few-shot scene classification, which aims to solve the problems of lacking label information and complex contextual semantics of remote sensing images. Experiments on three remote sensing datasets prove the efficacy of the proposed approach. Specifically, the hypergraph propagation module can take advantage of adjacent samples with similar semantics and improve the representation ability of class prototypes in the case of fewer samples. Moreover, the cross-layer prototype matching module is able to dynamically match features of different scales and angles, which can effectively predict pseudolabels with high confidence and enhance the accuracies of few-shot scene classification.

In our proposed method, a set metric method based on a hypergraph network is utilized to predict pseudolabels of unlabeled samples. This method will inevitably introduce false pseudolabels, resulting in confused information. In future work, it should study the more effective utilization of pseudolabels in semisupervised few-shot learning.

References

[1] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[2] L. Tian, X. Zhong, and M. Chen, "Semantic segmentation of remote sensing image based on GAN and FCN network model," *Sci. Program.*, vol. 2021, pp. 1–11, Nov. 2021.

[3] Z. Zakria, J. Deng, R. Kumar, M. S. Khokhar, J. Cai, and J. Kumar, "Multiscale and direction target detecting in remote sensing images via modified YOLO-v4," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1039–1048, 2022.

[4] J. Geng, W. Ma, and W. Jiang, "Causal intervention and parameter-free reasoning for few-shot SAR target recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12702–12714, Dec. 2024.

[5] A. Afrasiyabi, H. Larochelle, J.-F. Lalonde, and C. Gagne, "Matching feature sets for few-shot image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9014–9024.

[6] J. Geng, B. Xue, and W. Jiang, "Foreground–Background contrastive learning for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5614112.

[7] A. H. H. Ngu, Q. Z. Sheng, D. Q. Huynh, and R. Lei, "Combining multi-visual features for efficient indexing in a large image database," *VLDB J.*, vol. 9, no. 4, pp. 279–293, Apr. 2001.

[8] R. Benmokhtar, B. Huet, and S.-A. Berrani, "Low-level feature fusion models for soccer scene classification," in *Proc. IEEE Int. Conf. Multimedia Expo*, Jun. 2008, pp. 1329–1332.

[9] O. A. B. Penatti, K. Nogueira, and J. A. dos Santos, "Do deep features generalize from everyday objects to remote sensing and aerial scenes domains?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 44–51.

[10] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "MARTA GANs: Unsupervised representation learning for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2092–2096, Nov. 2017.

[11] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[12] X. Sun, B. Wang, Z. Wang, H. Li, H. Li, and K. Fu, "Research progress on few-shot learning for remote sensing image interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2387–2402, 2021.

[13] L. Fei-Fei, R. Fergus, and P. Perona, "One-shot learning of object categories," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 4, pp. 594–611, Apr. 2006.

[14] X. Luo et al., "Rectifying the shortcut learning of background for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 13073–13085.

[15] R. Kwitt, S. Hegenbart, and M. Niethammer, "One-shot learning of scene locations via feature trajectory transfer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 78–86.

[16] H. Zhang, J. Zhang, and P. Koniusz, "Few-shot learning via saliency-guided hallucination of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2019, pp. 2765–2774.

[17] A. Bär, N. Houlsby, M. Dehghani, and M. Kumar, "Frozen feature augmentation for few-shot image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16046–16057.

[18] D. Kang, H. Kwon, J. Min, and M. Cho, "Relational embedding for few-shot classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Dec. 2021, pp. 8802–8813.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.

[19] C. Zhang, Y. Cai, G. Lin, and C. Shen, "DeepEMD: Few-shot image classification with differentiable Earth mover's distance and structured classifiers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12200–12210.

[20] D. Wertheimer, L. Tang, and B. Hariharan, "Few-shot classification with feature map reconstruction networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8012–8021.

[21] C. Zhang, Y. Cai, G. Lin, and C. Shen, "DeepEMD: Differentiable Earth mover's distance for few-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5632–5648, May 2023.

[22] X. Chen, N. Rostamzadeh, B. N. Oreshkin, and P. O. Pinheiro, "Adaptive cross-modal few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, Mar. 2019, pp. 4847–4857.

[23] H. Zhang, J. Xu, S. Jiang, and Z. He, "Simple semantic-aided few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28588–28597.

[24] A. Qin et al., "Few-shot learning with prototype rectification for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409015.

[25] J. Zheng, M. Li, X. Li, P. Zhang, and Y. Wu, "Revisiting local and global descriptor-based metric network for few-shot SAR target classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5205814.

[26] X. Wang, X. Wang, B. Jiang, and B. Luo, "Few-shot learning meets transformer: Unified query-support transformers for few-shot classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 12, pp. 7789–7802, Dec. 2023.

[27] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, Aug. 2017, pp. 1126–1135.

[28] Z. Li, F. Zhou, F. Chen, and H. Li, "Meta-SGD: Learning to learn quickly for few-shot learning," 2017, *arXiv:1707.09835*.

[29] F. Zhou, B. Wu, and Z. Li, "Deep meta-learning: Learning to learn in the concept space," 2018, *arXiv:1802.03596*.

[30] Y. Chen, Z. Liu, H. Xu, T. Darrell, and X. Wang, "Meta-baseline: Exploring simple meta-learning for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9042–9051.

[31] S. Baik, M. Choi, J. Choi, H. Kim, and K. M. Lee, "Learning to learn task-adaptive hyperparameters for few-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1441–1454, Mar. 2024.

[32] H.-J. Ye, L. Ming, D.-C. Zhan, and W.-L. Chao, "Few-shot learning with a strong teacher," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1425–1440, Mar. 2024.

[33] K. Huang, J. Geng, W. Jiang, X. Deng, and Z. Xu, "Pseudo-loss confidence metric for semi-supervised few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 8671–8680.

[34] X.-S. Wei, H.-Y. Xu, Z. Yang, C.-L. Duan, and Y. Peng, "Negatives make a positive: An embarrassingly simple approach to semi-supervised few-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 2091–2103, Apr. 2024.

[35] Y. Wang, C. Xu, C. Liu, L. Zhang, and Y. Fu, "Instance credibility inference for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12833–12842.

[36] P. Rodríguez, I. Laradji, A. Drouin, and A. Lacoste, "Embedding propagation: Smoother manifold for few-shot classification," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Aug. 2020, pp. 121–138.

[37] Y. Liu et al., "Learning to propagate labels: Transductive propagation network for few-shot learning," 2018, *arXiv:1805.10002*.

[38] Y. Feng, H. You, Z. Zhang, R. Ji, and Y. Gao, "Hypergraph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 3558–3565.

[39] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[40] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[41] J. Feng, L. C. Jiao, X. Zhang, and D. Yang, "Bag-of-visual-words based on clonal selection algorithm for SAR image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 691–695, Jul. 2011.

[42] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7844–7853, Sep. 2021.

[43] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[44] Q. Zeng and J. Geng, "Task-specific contrastive learning for few-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 143–154, Sep. 2022.

[45] Q. Zeng, J. Geng, W. Jiang, K. Huang, and Z. Wang, "IDLN: Iterative distribution learning network for few-shot remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] J. Ling, L. Liao, M. Yang, and J. Shuai, "Semi-supervised few-shot learning via multi-factor clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14544–14553.

[47] M. Lazarou, T. Stathaki, and Y. Avrithis, "Iterative label cleaning for transductive and semi-supervised few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8731–8740.



Jie Geng (Member, IEEE) received the B.S. degree in electronic and information engineering from Dalian University of Technology, Dalian, China, in 2013, and the Ph.D. degree from the School of Information and Communication Engineering, Dalian University of Technology, in 2018.

He is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include SAR image processing, remote sensing image classification, and few-shot learning.



Ran Chen received the bachelor's degree from Hunan University, Changsha, China, in 2021, and the master's degree from Northwestern Polytechnical University, Xi'an, China, in 2024, where he is currently pursuing the Ph.D. degree with the School of Electronics and Information.

His research interests include remote sensing image scene classification and few-shot learning.



Bohan Xue received the master's degree from Northwestern Polytechnical University, Xi'an, China, in 2024.

His research interests include remote sensing image scene classification and few-shot learning.



Wen Jiang received the bachelor's and master's degrees from PLA Information Engineering University, Zhengzhou, China, in 1994 and 1997, respectively, and the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2009.

She is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include information fusion, artificial intelligence, remote sensing image processing, and intelligent algorithm security.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:16:03 UTC from IEEE Xplore. Restrictions apply.